

Package Insert I. Brand or Product Name: ENROLIN-LD 200MG/ML ORAL SOLUTION II. Name and Strength of Active Substance: Enrofloxacin Base 200mg in 1mL Preservative : Benzyl Alcohol 1% III. Product Description: A yellow color, clear solution. IV. Target Species: Poultry (broiler chickens and turkeys) and rabbits.. V. Pharmacodynamics: Enrofloxacin is a synthetic fluoroquinolone antimicrobial agent. Mechanism of action: Two essential enzymes in DNA replication and transcription have been identified as the molecular targets of fluoroquinolones: DNA gyrase, Topoisomerase II and Topoisomerase IV. In Gram positive bacteria, the main target would be Topoisomerase IV instead of Topoisomerase II. Enrofloxacin exerts its bactericidal effect by interacting with the A subunit of bacterial DNA gyrase (enzyme responsible for controlling supercoiling of bacterial DNA during replication), preventing negative axial rotation of the DNA molecule. The closure of the standard double helix is inhibited, causing irreversible degradation of chromosomal DNA and blocking the process of replication, transcription and recombination of bacterial DNA. This mechanism of action is concentration dependent. Fluoroquinolones also act on bacteria in the stationary phase by altering the permeability of the cell membrane. Antibacterial spectrum: Enrofloxacin is effective against Gram negative and Gram positive bacteria and <i>Mycoplasma</i> spp. In vitro susceptibility has been demonstrated in strains of (i) Gram negative species such as <i>Pasteurella multocida</i> and <i>Avibacterium (Haemophilus) paragallinarum</i> and (ii) <i>Mycoplasma gallisepticum</i> and <i>Mycoplasma synoviae</i>. Types and Mechanisms of resistance: Five mechanisms of resistance to fluoroquinolones have been described: (i) point mutation of the genes encoding for DNA gyrase and/or topoisomerase IV causing alterations in the respective enzymes, (ii) alterations of drug permeability in Gram-negative bacteria, (iii) expulsion mechanisms, (iv) plasmid mediated resistance, and (v) gyrase protective proteins. These mechanisms cause a lower sensitivity of bacteria to fluoroquinolones. Cross-resistances among antimicrobials of the fluoroquinolone class are frequent. VI. Pharmacokinetics: Enrofloxacin has a relatively high oral bioavailability in almost all species studied. After oral administration of enrofloxacin to chickens and rabbits the maximum concentration is reached between 0.5 to 2.5hours. The maximum concentration, after an administration of a therapeutic dose is between 1 to 2.5ug/ml. Concomitant administration of compounds containing multivalent cations (antacids, milk or milk replacers) decreases the oral bioavailability of	fluoroquinolones. Fluoroquinolone are characterized by extensive diffusion into body fluids and tissues, in some reaching concentrations higher than those found in plasma. In addition, they are widely distributed in skin, bone and semen, also reaching the anterior and posterior chambers of the eye. They cross the placenta and the blood brain barrier. They are also stored in phagocytic cells (alveolar macrophages, neutrophils) and therefore are effective against intracellular microorganisms. The degree of metabolism varies between species and is around 50 to 60%. The biotransformation of enrofloxacin in the liver gives rise to an active metabolite, ciprofloxacin. In general, metabolism occurs through processes of hydroxylation and oxidation to oxofluoroquinolones. Other reactions that also occur are N-dealkylation and conjugation with glucuronic acid. Excretion occurs through the bile and kidney, the latter being the predominant one. Renal excretion is carried out by glomerular filtration, and also by active tubular secretion through the organic anion pump. VII. TO BE PRESCRIBED BY REGISTERED VETERINARY SURGEONS ONLY Indication: Treatment of infections caused by bacteria sensitive to enrofloxacin: Chickens: <i>Mycoplasma gallisepticum</i>, <i>Mycoplasma synoviae</i>, <i>Avibacterium paragallinarum</i>, and <i>Pasteurella multocida</i>. Turkeys: <i>Mycoplasma gallisepticum</i>, <i>Mycoplasma synoviae</i>, and <i>Pasteurella multocida</i>. Rabbits: Treatment of respiratory infections caused by strains of <i>P. multocida</i> sensitive to enrofloxacin. VIII. Recommended Dose and Administration: Route of administration: administration in animals' drinking water. Chickens and Turkeys: 10mg enrofloxacin per kg bodyweight per day, for 3 to 5 consecutive days (equivalent to 0.05ml of Product per kg bodyweight per day) Treatment for 5 consecutive days in case of mixed infections and in chronic progressive forms. If no clinical improvement is observed within 2 to 3 days, alternative antimicrobial therapy should be considered based on sensitivity tests. Rabbits: 10mg enrofloxacin per kg bodyweight per day for 5 days (equivalent to 0.05ml of Product per kg bodyweight per day). The weight of the animals should be determined as accurately as possible to avoid underdosing. Water consumption depends on the clinical situation of the animal and the time of the year. To ensure a correct dosage, the concentration of enrofloxacin in the water will be adjusted taking into account the daily consumption. Based on the recommended dose, the number and weight of the animals to be treated, the exact daily dose of the drug should be calculated using the following formula: $\frac{10\text{mg/kg/day} \times \text{mean weight of animals}}{\text{ml of Product/L of water} = 200\text{mg/ml} \times \text{mean water consumption (L/day)}}$ Medicated water should be prepared every 24hours, immediately before starting treatment. Medicated water should be the only source of water during the entire treatment.	Check that the administered dose is completely consumed by checking the delivery systems periodically. Before starting the treatment, empty the supply system and fill with the medicated water. Use appropriate and properly calibrated dosing equipment. IX. Contraindication: Do not use for prophylaxis. Do not use if resistance or cross-resistance to (fluoro)quinolones has been identified in the group of birds to be treated. Do not use in cases of hypersensitivity to enrofloxacin, other (fluoro)quinolones or to any of the excipients. X. Warning and Precautions: <u>Special warnings for each target species:</u> Treatment of infections by <i>Mycoplasma</i> spp. may not eradicate the microorganism. <u>Special precautions for use in animals:</u> The drug should be used taking into account official and local policies on antimicrobials. Fluoroquinolones should be reserved for the treatment of clinical conditions that have responded poorly or are expected to respond poorly to other groups of antimicrobials. To the extent possible, the use of fluoroquinolones should always be based on sensitivity analyses. The use of the drug without following the instructions included in the data sheet may increase the prevalence of bacteria resistant to the fluoroquinolones and reduce the effectiveness of treatment with other quinolones due to possible cross resistance. Since enrofloxacin was first licensed for use in poultry, there has been a general reduction in susceptibility of <i>E.coli</i> to fluoroquinolones and the emergence of resistant organisms. Resistance has also been reported in <i>Mycoplasma synoviae</i> in the EU. <u>Special precautions to be taken by the person administering the veterinary medicinal product to animals:</u> People with known hypersensitivity to (fluoro)quinolones or to any of the excipients should avoid all contact with the veterinary medicinal product. Avoid contact with skin and eyes. Wear gloves and handle the veterinary medicinal product carefully to avoid contact during its incorporation into drinking water. In case of contact, wash immediately and thoroughly with plenty of water. If symptoms appear after exposure, such as skin rash, consult a doctor and display these warnings. Swelling of the face, lips or eyes or respiratory distress are more serious signs that require urgent medical attention Do not smoke, eat or drink while handling the veterinary medicinal product.. XI. Interaction with other medicinal product: In vitro, an antagonistic effect has been demonstrated when fluoroquinolones are combined with other bacteriostatic antimicrobials such as macrolides or tetracyclines and phenicols. Concomitant administration of substances containing magnesium or aluminium may reduce the absorption of enrofloxacin. XII. Pregnancy and Lactation: Its use is not authorized in laying birds whose eggs are used for human consumption. Do not use in replacement laying birds in the 14 days prior to the start of laying.
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- XIII. Side Effects:
None known.
- XIV. Symptoms and Treatment of Overdose:
In the event of overdose, the symptoms would consist of weak stimulation of spontaneous motility, so treatment should be discontinued. Fluoroquinolone poisoning can cause nausea, vomiting and diarrhea. The use of fluoroquinolones during the growth phase together with a significant and prolonged increase in the intake of drinking water and consequently of enrofloxacin, possibly due to high temperatures, maybe associated with damage to the articular cartilage.
- XV. Withdrawal Period:
Chickens (meat) : 7 days
Turkeys (meat) : 13 days
Rabbits (meat) : 2 days
Not permitted for use in laying birds producing eggs for human consumption. Do not administer to replacement laying birds in the 14 days prior to the start of laying.
- XVI. Storage Condition:
Stored at ≤ 30°C / Store below 30°C.
Protect from light.
KEEP OUT OF REACH OF CHILDREN.
JAUHKAN DARI KANAK-KANAK.
- XVII. Proposed Shelf Life:
As packaged for sale: 3 years
After first opening of the immediate packaging: 28 days
After dilution or reconstitution: 24 hours
- XVIII. Dosage Forms and packaging available:
In solution form, packed in 1L and 25L plastic containers.
- XIX. Name and Address of manufacturer
NAM PHARMA SDN. BHD.
5, Lebuhr Perusahaan Klebang 11,
Taman Perindustrian Antarabangsa IGB,
31200 Chemor, Perak, Malaysia.
- XX. Name and Address Product holder
KTY CHEMICALS SDN BHD
65, Jalan Cenderawasih 3,
Taman Cenderawasih Indah,
14300 Nibong Tebal, SPS, Penang, Malaysia.
- XXI. Date of Revision of Package Insert:
21.9.2023